INCH-POUND

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DETAIL SPECIFICATION

TRANSMITTER, PRESSURE TRU-81/A, OIL, 0-500 PSI

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE. This specification covers one type of remote-indicating, 320 degree movement, synchro style, differential, oil-pressure transmitter, designated TRU-81/A.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@DLA.MIL. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-DTL-6856	- Indicator, Pressure, Synchro, 1-1/2 Inch.
MIL-I-7057	- Indicators, Synchro, Aircraft, General Specification for.
MIL-A-8625	- Anodic Coatings for Aluminum and Aluminum Alloys.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-461	- Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment.
MIL-STD-810	- Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1916	- DoD Preferred Methods for Acceptance of Products.
MS33678	- Connector, Receptacle, Electric, Integral Mounting.

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-781	- Reliability Test Methods, Plans, and Environments for Engineering, Development, Qualification, and Production.
MIL-HDBK-831	- Preparation of Test Reports
MIL-HDBK-5400	- Electronic Equipment, Airborne, General Guidelines for.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Specifications, Standards, and handbooks (specific revision). The following specifications, standards, and handbooks of the exact revision listed below form a part of this specification to the extent specified herein.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-704A	- Aircraft Electric Power Characteristics. (dated 9 August 1966)
MIL-STD-810C	- Environmental Engineering Considerations and Laboratory Tests. (dated 10 March 1975, Vibration tests only)

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

SAE INTERNATIONAL

SAE-AMS-QQ-S-763	- Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings (DoD adopted).
SAE-AS8879	- Screw Threads - UNJ Profile, Inch Controlled Radius Root with Increased Minor Diameter (DoD adopted).
SAE-AS20708	- Synchros, General Specification for (DoD adopted).
SAE-AS5202	- Port or Fitting End, Internal Straight Thread, Design Standard (DoD adopted).

(Copies of these documents are available online at http://www.sae.org/ or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM21209	- Insert, Screw Thread, Coarse and Fine, Screw Locking, Helical Coil, Cres (DoD adopted).

(Copies of these documents are available online at http://www.aia-aerospace.org/ or from the Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209.)

AMERICAN BEARING MANUFACTURERS ASSICIATION (ABMA)

ABMA 12.1	- Instrument Ball Bearings, Metric Design
ABMA 12.2	- Instrument Ball Bearings, Inch Design

(Copies of these documents are available online at http://www.abma-dc.org/ or from American Bearing Manufacturers Association, 2025 M Street, NW, Suite 800, Washington, DC 20036.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Transmitters furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award (see 4.3 and 6.3).

3.2 Materials. Materials used in the manufacture of transmitters shall be suitable for the purpose and shall conform to applicable government specifications. Materials conforming to

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contractor's specifications may be used provided such specifications are approved by the government and contain provisions for adequate tests. The use of the contractor's specifications will not constitute waiver of government inspection.

3.2.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them.

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.2.3 Metals. Metals shall be of the corrosion resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.2.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other.

3.2.5 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

3.2.6 Corrosive fumes. Materials used in the construction of the transmitter shall not produce corrosive or deleterious fumes under the test conditions specified herein.

3.2.7 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings that will crack, chip, or scale with age or the extremes of climatic and environmental conditions shall not be used.

3.3 Design and construction.

3.3.1 Design. The transmitter shall be designed to operate with indicators conforming to MIL-DTL-6856 and MIL-I-7057.

3.3.1.1 Outline dimension. The outline dimensions of the transmitter shall conform to figure 1.

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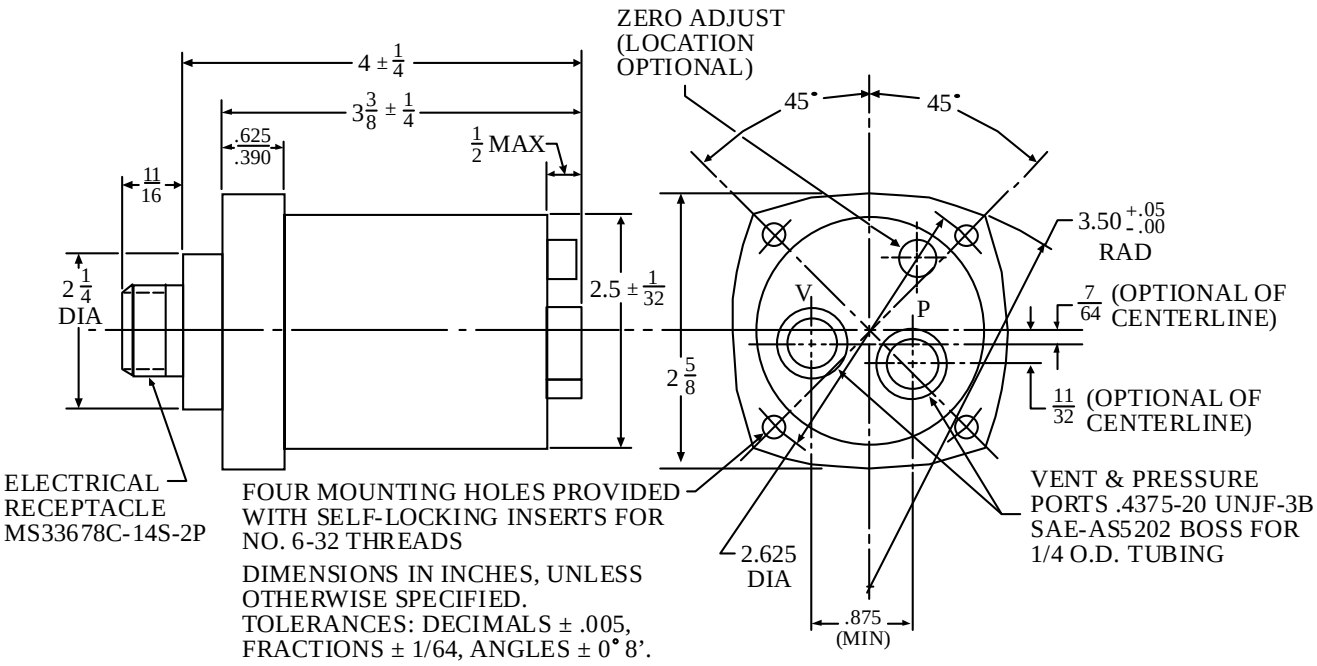


FIGURE 1. Case dimensions.

3.3.1.2 Adjustments. The transmitter shall be provided with a means for adjusting or correcting indications that shall be simple to manipulate and such that adjustments can be accomplished with tools ordinarily available. The means of adjustment shall be accessible from the exterior of the case and shall provide a minimum of ±7 angular degrees of synchro rotor adjustment from electrical zero.

3.3.1.3 Mechanisms. Transmitter mechanisms shall be comprised of units distinct from their cases. The units shall be sufficiently rugged so that small distortions of the cases will not affect the accuracy of the indications.

3.3.1.4 Adjustment locks. All calibration adjustments shall be provided with securing devices to lock the adjusting mechanism in place to prevent the adjustment from drifting from its original setting due to environmental conditions specified herein. Provisions for locking shall be made on the calibration adjustment so that it can not be readily moved.

3.3.1.5 Reliability. The transmitter shall have specified mean time between failures (MBTF) of 2,000 hours when tested as specified in 4.6.23.1.

3.3.1.6 Lubricant. The contractor shall select a lubricant or lubricants that are suitable for use on the gears, joints, bearings, and pinions of the transmitter. Prior to selecting the lubricant or lubricants, the contractor shall make an investigation relative to the lubricant most suitable for the application with due consideration being given to the environmental conditions under which the transmitter must operate.

3.3.1.7 Synchro transmitters. The synchro transmitter component of the pressure transmitter shall meet all performance requirements of SAE-AS20708 except as specified herein.

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3.3.1.7.1 Synchro bearings. The precision ball bearings used in the synchro shall be corrosion resistant alloy steel (stainless steel) conforming to SAE-AMS-QQ-S-763, class 440C, and shall be made to ABEC tolerance 5P per ABMA 12.1 or 12.2 as applicable.

3.3.1.7.2 Electrical zero. A means shall be provided for readily setting the synchro at electrical zero. The synchro transmitter shall be designed so that the synchro electrical zero position can be set and locked securely so that it will remain in that position for the service life of the transmitter or until reset.

3.3.1.8 Overpressure stop. An overpressure stop shall be provided to restrain the prime mover. The stop shall permit the transmitter rotor to rotate 12 degrees to ±3 degrees beyond the 500 pounds per square inch (psi) position. The stop shall not be damaged by the application of 750 psi.

3.3.1.9 Gears, pivots, joints, and bearings. The gears, pivots, joints, and bearings shall resist wear, corrosion, and friction when interfacing with one another. The interfacing materials shall not be dissimilar as described in MIL-STD-889.

3.3.1.9.1 Large area interface surface. The gears, pivots, joints, and bearings shall have substantially large interface surfaces for the purpose of minimizing the effects of wear upon the calibration and accuracy of the transmitter. Single point type surfaces, such as attained by pivots made of two perpendicular cylinders, will not be acceptable. Interface surfaces made up of two parallel cylinders or curved surfaces will be acceptable if sufficient interface area is provided to minimize the effects of wear.

3.3.1.10 Attitude. The transmitter shall be so designed that it will operate as specified herein in any position.

3.3.1.11 Connections. The transmitter shall have a pressure port connection and separate vent port connection as shown in figure 1.

3.3.1.12 Containment. There shall be no flow path between the two port connections or either port connection and the outside of the transmitter. Except for the pressure and vent ports, no single transmitter failure shall result in such a path.

3.3.2 Construction. The transmitter shall be constructed so that no parts will work loose in service. It shall be built to withstand strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. Pivots, bearings, and gears shall not bind or shake and shall be as nearly frictionless as possible.

3.3.2.1 Mounting provisions. The mounting flange of the transmitter shall be constructed in accordance with figure 1. The four mounting holes shall be provided with screw locking helical coils in accordance with NASM21209 inserted from the electrical connector side of the transmitter. The helical coils shall accept and lock #6-32 threaded screws.

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3.4 Performance. The transmitter shall meet the performance requirements specified herein and operate within the tolerances specified in table I.

TABLE I. Scale error tolerances.

Scale point psi	Deg[1]	Room temperature, friction, vibration, sensor match	During/after reliability, high and low temp exposure, overpressure	High and low temperature	Pointer oscillation	Position error
0	20	3.0	4.5	3.75	2.25	
50	52	3.5	5.25	4.5	2.75	
100	84	4.0	6.0	5.0	3.0	
150	116	4.5	6.75	5.5	3.5	
200	148	5.0	7.5	6.25	3.75	
250	180	5.5	8.25	7.0	4.25	5.0 (at mid scale)
300	212	6.0	9.0	7.5	4.5	
400	276	7.0	10.5	8.75	5.25	
500	340	8.0	12.0	10.0	6.0	

[1]Assumes indicating standard electrical zero in accordance with SAE-AS20708.

3.4.1 Temperature. The transmitter shall operate at temperatures ranging from -65 ºF to 329 ºF (-54 ºC to +165 ºC) and exposure to temperatures ranging from -79 ºF to +450 ºF (-62 ºC to +232 ºC).

3.4.2 Vibration. The transmitter shall withstand vibration of 20 gravity (g) peak for three hours on each axis.

3.4.3 Altitude-temperature. The transmitter shall withstand absolute pressure equivalent to an 85,000-foot altitude at -65 ºF (-54 ºC).

3.4.4 Salt fog. The transmitter shall withstand exposure to salt sea atmosphere for 48 hours.

3.4.5 Overpressurization

3.4.5.1 Overpressure (750 psi). The transmitter shall withstand an overpressure of 750 psi for a 10 minute period.

3.4.5.2 Overpressure (1000 psi). The transmitter with a compromised case shall withstand an overpressure of 1000 psi without leakage or failure.

3.4.6 Dielectric strength. The transmitter shall withstand 200 volts (Vac) at a commercial frequency applied between any pin of the electrical connector not connected to the case and the case for a minimum period of five seconds.

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3.4.7 Voltage and frequency. The transmitter shall withstand voltage and frequency variations from 22 to 30 Vac and 380 to 420 hertz (Hz), or any combinations thereof.

3.4.8 Electromagnetic interference. Unless otherwise specified (see 6.2), the transmitter shall meet the electromagnetic interference requirements of MIL-STD-461 for Air Force aircraft.

3.4.9 Damping. When the transmitter is subjected to an applied oil pressure sufficient to produce an indication of 500 psi and the pressure is released from 500 to 100 psi within 0.1 seconds, the corresponding time required for the transmitter indication to change from 500 to 100 psi shall be between 3 and 7 seconds. There shall be no damage to the instrument as a result of the pressure change.

3.4.10 Indicator operation. The transmitter shall be capable of operating either one or two indicators with size 10 synchros without a change in output equivalent to more than 1 degree of arc.

3.4.11 Pressure bosses and electrical connector. The pressure bosses and electrical connector shall withstand an applied force of 25 foot-pounds bending movement with no evidence of damage to welds or soldering, distortion of base plates, and without the calibration shifting out of tolerance.

3.4.12 Scale error. Scale error shall not exceed the tolerances specified in table I.

3.4.13 Friction and position error. Friction and position error shall not exceed the tolerances specified in table I.

3.4.14 Effect of lead length. When the lead length (No. 20 American Wire Gauge (AWG) stranded copper wire) between the transmitter and indicator synchro stators is changed from 5 to 300 feet, the indication shall not change more than 1 degree of arc.

3.4.15 Case leakage. When a pressure of approximately 0 psi is applied to the exterior of the transmitter case, the leak rate shall not exceed 0.10 micron cubic foot per hour when 10 percent helium is used in the filling medium and shall not exceed 1.0 micron cubic foot per hour when 100 percent helium is used as the filling medium.

3.4.16 Sensors matching. The transmitter will remain accurate when pressure is applied to the vent port.

3.5 Case. The case shall conform to figure 1 and shall provide a hermetically sealed enclosure for the internal mechanism of the transmitter. Sealing of the case shall be accomplished so that the seal will not be dependent upon material that would be affected by the action of any atmosphere to which the transmitter may be subjected.

3.5.1 Filling medium. The filling medium shall be at least 98 percent free of dust particles and shall contain not more than 0.006 milligram of water vapor per liter (dew point -65 ºC) at the filling pressure. The filling medium shall be either a mixture of 88 to 92 percent nitrogen and the remainder helium, or 100 percent helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

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3.5.2 Case pressure release. A blowout device shall be provided for releasing pressure in the case. The transmitter shall be designed and constructed so that if the pressure sensor connected to the pressure port leaks, the pressure in the case will be relieved and the escaping fluid will be discharged through the vent port.

3.6 Electronic components. Electronic components and the application thereof should be in accordance with MIL-HDBK-5400. Nonstandard parts shall not be used except with prior approval of the procuring activity.

3.7 Electrical connector. An MS33678C-14S-2P hermetically sealed electrical connector shall be provided.

3.8 Electrical characteristics. The transmitter synchro shall meet requirements of 3.8.1 through 3.8.4 with normal excitation of 26 Vac 400 Hz single-phase power applied to the primary (rotor) from a source having electrical characteristics conforming to MIL-STD-704A (9 August 1966) for category B equipment.

3.8.1 Open secondary. The current drawn by the primary of the self-synchronous motor shall not exceed 240 milliamperes (ma) with secondaries (stator) open circuited.

3.8.2 Power consumption. Power consumption of the transmitter synchro shall not exceed 6.24 Volt-amperes.

3.8.3 Secondary voltage. The open-circuit voltage measured between any two secondary leads shall be 11.8 ± 0.3 Vac. The variations between the three maximum secondary voltages on any one synchro shall not exceed 0.2 Vac.

3.8.4 Power supply variation. The transmitter shall operate with a supply voltage between 22 and 30 Vac at a frequency between 380 and 420 Hz.

3.9 Wiring.

3.9.1 External. The electrical receptacle shall be so wired that the 26 Vac 400 Hz power supply will be connected to terminals A and B. External wiring between the transmitter and indicator shall be as shown in figure 2.

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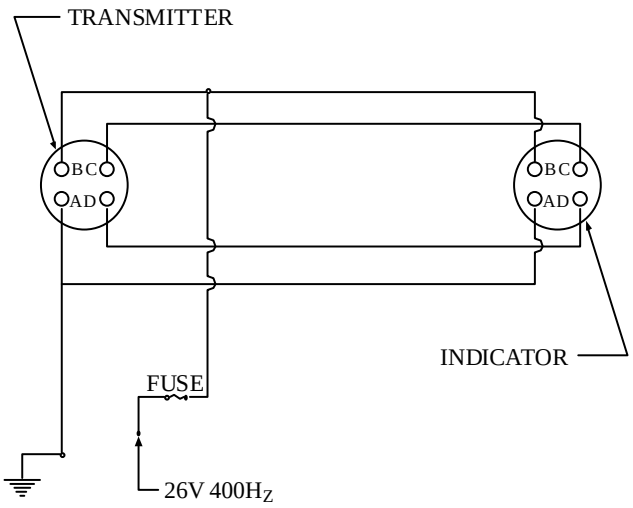


FIGURE 2. Wiring diagram - external, single indicator.

3.9.2 Internal. Internal wiring of the transmitter shall be color coded and shall be connected to the receptacle pins as shown in figure 3. When gears are used to drive the transmitter counterclockwise for increasing values of the parameter being measured, terminals S1 and S3 shall be reversed internally to cancel the effects of the gear reversal to ensure clockwise rotation of the indicator synchro shaft and pointer with increasing values.

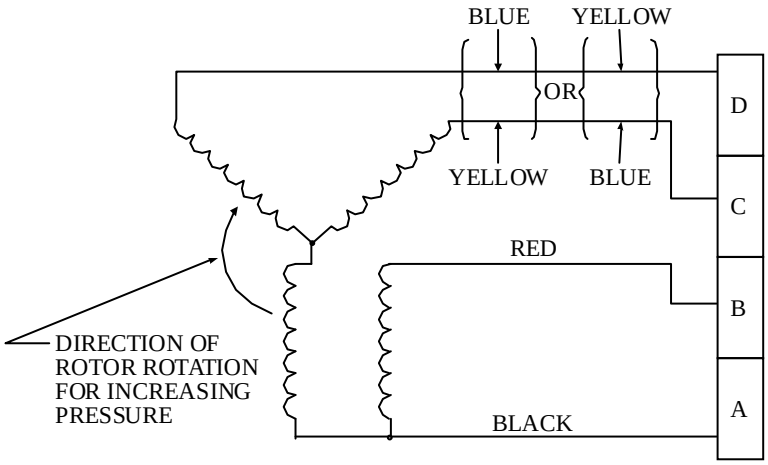


FIGURE 3. Internal electrical circuit.

3.10 Finishes and protective coatings.

3.10.1 Aluminum alloy parts. Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625, except the small holes, straight threads, case inserts, and parts in the sealed enclosure. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541.

3.10.1.1 Abrasion resistance. When abrasion resistance is a factor, chemical film in accordance with MIL-DTL-5541 shall not be used in lieu of anodizing.

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3.11 Weight. The weight of the transmitter shall not exceed 1.7 pounds.

3.12 Screw threads. Unless otherwise specified herein, the threads of all machine screws shall conform to SAE-AS8879.

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.13.1 Nameplate. A nameplate shall be permanently attached to the rear of the case or the required information shall be permanently etched on the side of the case where paint is not needed as protective coating. Where applicable, the information marked thereon shall conform to MIL-STD-130. The name plate shall not be glued or cemented to the case nor shall metal decals be used. Etching shall be permanent so that it will withstand all normal environments for the life of the transmitter.

3.14 Cadmium. Cadmium is not to be used in the transmitter unless approved by the preparing activity. There shall be no cadmium used on the outside case of the transmitter. The fluid whose pressure is being measured by the transmitter shall not come in contact with any cadmium in the transmitter.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.5).

4.2 Inspection condition. Unless otherwise specified, all inspections shall be performed in accordance with test conditions specified in the applicable test method document or applicable paragraph(s) in this specification (see 6.2).

4.2.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, the test shall be conducted at an atmospheric pressure of 29.92 inches of mercury (Hg) and at an ambient temperature of 329 ºF + 9 ºF - 0 ºF (165 ºC + 5 ºC - 0 ºC). For tests conducted at room temperature, the temperature shall be 77 ºF ± 9 ºF (25 ºC ± 5 ºC).

4.2.2 Test voltage. The transmitter shall be tested with 26 Vac 400 Hz single phase power. Unless otherwise specified, the voltage and frequency shall be maintained with ±1 percent of the nominal value.

4.2.3 Tapping. The transmitter shall be lightly tapped or vibrated before each scale error test reading.

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4.2.4 Test standard. The transmitter shall be tested against an indicating standard having maximum error of ±0.1 circular degree. The pressure source, such as a precision deadweight tester, shall have a maximum inherent error of ±0.05 percent.

4.3 Qualification inspection. The test samples shall consist of ten transmitters representative of the production process. Three of the samples shall undergo all of the tests specified in section 4.6 excluding reliability and the remaining seven shall be subjected to the reliability test in section 4.6.23.1 only.

4.4 Maintenance of qualification. To sustain qualification status, the manufacturer shall contact the qualifying activity at two-year intervals. The qualifying activity will establish the initial contact date and request the appropriate documentation from the manufacturer to demonstrate that the capability to produce the items is still available. Failure to contact the qualifying activity within 30 days after the end of each two year period may result in the loss of qualification.

4.5 Conformance inspection. Conformance inspection shall be in accordance with MIL-STD-1916 and shall consist of the individual tests (see 4.5.1) for each unit produced, as well as sampling plan A tests (see 4.5.2) or sampling plan B tests (see 4.5.3), if required by the procuring activity (see 6.2).

a. Individual tests (see 4.5.1).
b. Sampling tests (see 4.5.2 or 4.5.3).

4.5.1. Individual tests. Each transmitter shall be subjected to the following tests as described in 4.6.

a. Examination of product (see 4.6.1).
b. Scale error (see 4.6.2).
c. Friction error (see 4.6.3).
d. Position error (attitude) (see 4.6.4).
e. Dielectric strength (see 4.6.5).
f. Case leakage (see 4.6.6).

4.5.2 Sampling plan A tests. Sampling plan A tests are in addition to the individual tests. The manufacturer shall select the type of sampling plan (i.e. attribute, variable, or continuous) in accordance with MIL-STD-1916. The sample size shall be selected in accordance with verification level I of MIL-STD-1916.

a. Individual tests (see 4.5.1).
b. Low temperature (see 4.6.8).
c. High temperature (see 4.6.9).
d. Vibration (see 4.6.10).
e. Electrical characteristics (see 4.6.11).
f. Damping (see 4.6.21).
g. Sensors matching (see 4.6.24).

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4.5.3 Sampling plan B tests. Sampling plan B tests are in addition to the individual and the sampling plan A tests. The manufacturer shall select the type of sampling plan (i.e. attribute, variable, or continuous) in accordance with MIL-STD-1916. The sample size shall be selected in accordance with verification level I of MIL-STD-1916.

a. Sampling plan A tests (see 4.5.2).
b. Low temperature exposure (see 4.6.12).
c. High temperature exposure (see 4.6.13).

4.6 Requirements cross-reference. Table II provides a cross-reference of section 3 requirements and section 4 verification testing.

TABLE II. Requirement-verification cross-reference.

Requirement paragraph	Test	Verification paragraph
3.4.12	Scale error	4.6.2
3.4.13	Friction error	4.6.3
3.4.13	Position error	4.6.4
3.4.6	Dielectric strength	4.6.5
3.4.15	Case leakage	4.6.6
3.3.1.7.2	Electrical zero and verification of terminal leads	4.6.7
3.4.1	Low temperature	4.6.8
3.4.1	High temperature	4.6.9
3.4.2	Vibration	4.6.10
3.8	Electrical characteristics	4.6.11
3.4.1	Low temperature exposure	4.6.12
3.4.1	High temperature exposure	4.6.13
3.4.3	Altitude temperature	4.6.14
3.4.8	Electromagnetic interference	4.6.15
3.4.4	Salt fog	4.6.16
3.4.7	Voltage and frequency	4.6.17
3.4.14	Effect of lead length	4.6.18
3.4.11	Pressure bosses and connector strength	4.6.19
3.3.1.8, 3.4.5.1	Overpressure (750 psi)	4.6.20
3.4.5.2	Overpressure (1,000 psi)	4.6.20.1
3.5.2	Case pressure release	4.6.20.2
3.4.9	Damping	4.6.21
3.5.1	Filling medium	4.6.22
3.3.1.5	Reliability test	4.6.23.1
3.4.16	Sensors matching	4.6.24

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, dimensions, and weight.

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4.6.2 Scale error. The transmitter shall be tested for scale error at both normal operating and room temperatures at the test points specified in table I for both up scale and down scale readings. The test points on the up scale reading shall be measured by bringing the input reading up to, but not exceeding, the specified test point. The difference in the scale error after vibration from the input value shall be brought down to, but not below, the specified test point.

4.6.3 Friction error. This test shall be conducted at both normal operating and room temperatures and may be performed concurrently with the scale error test specified in 4.6.2. The difference in reading before vibrating the transmitter and after vibrating shall be the friction error, which shall not exceed the tolerance specified in table I.

4.6.4 Position error. This test may be conducted with the scale error test specified in 4.4.2 at room temperature only. The transmitter shall be energized to obtain a reading of approximately one-half scale. The change in indication produced by rotating the transmitter from normal operating position to a position 90 degrees clockwise and counterclockwise (longitudinal axis vertical) shall be noted. The maximum change in indication shall not exceed the tolerance specified for position error in table I.

4.6.5 Dielectric strength. This test shall be conducted at room temperature. A potential 200 Vac root mean square (rms) at commercial frequency shall be applied between any pin of the electrical connector, not connected to the case and the case for a minimum period of five seconds. There shall be no insulation breakdown.

4.6.6 Case leakage. The transmitters shall be tested for leaks at room temperature by means of a mass-spectrometer-type helium leak detector. The detected leak rate for the 10 percent helium and 90 percent nitrogen filling medium shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 12 to 14.7 psi.

4.6.7 Electrical zero and verification of terminal leads. This test shall be conducted at room temperature. With power applied to the transmitter in accordance with figure 3, pressure shall be applied at the pressure connection of the transmitter with the vent connection open to ambient pressure. Electrical zero and verification of terminal leads shall be as follows:

a. The voltage from PIN C to PIN D (S3S1) shall begin at null and shall be in phase with rotor voltage PIN A and PIN B as the pressure is decreased from approximately 250 psig to 0 psig.
b. The voltage between PIN B and PIN C shall be near a minimum and shall decrease to minimum as the pressure is increased from 250 psig to 296 psig.
c. Increasing pressure 250 psig to 296 psig shall result in an increasing PIN C to PIN A voltage and decreasing PIN A to PIN D voltage.

4.6.7.1 Electrical zero adjustment. With pressure applied, the transmitter shall be subjected to and shall meet the test specified in 4.6.7, during which the zero adjustment shall be operated to the extreme positions. The required minimum adjustment shall be ±7 angular degrees from the electrical zero position.

4.6.8 Low temperature. The transmitter shall be subjected to a low temperature test in accordance with MIL-STD-810, method 502 at -65 ºF ± 4 ºF (-54 ºC ± 2 ºC) for a period of four

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hours after which it shall be subjected to the scale error test specified in 4.6.2 and the friction error test specified in 4.6.3 while still at that temperature. The scale error and friction error shall not exceed the tolerances specified in table I for low temperature.

4.6.9 High temperature. The transmitter shall be placed in a chamber and subjected to a temperature of 329 ºF + 9 ºF - 0 ºF (165 ºC + 5 ºC - 0 ºC) for a four hour period. Throughout this test, the transmitter shall be electrically connected to an appropriate indicator with power applied. At the end of the four hour period and with the transmitter still at the specified temperature, it shall be subjected to the scale error test specified in 4.6.2 and the friction error test specified in 4.6.3. The scale error and friction error shall not exceed the tolerances specified in table I.

4.6.10 Vibration. The transmitter shall be subjected to a vibration test in accordance with MIL-STD-810C (10 March 1975), method 514 equipment category (b.1), figure 514.2-2, curve L. At the conclusion of this test, the transmitter shall meet the requirements of the case leakage test specified in 4.6.6 and it shall be subjected to the scale error test specified in 4.6.2 and the friction error test specified in 4.6.3. The scale error and friction error shall not exceed the tolerances specified in table I for vibration.

4.6.11 Electrical characteristics. The transmitter shall be tested to determine compliance with the requirements specified herein with respect to primary current, power input, secondary voltage, and operation of either one or two indicators with a change in indication of not more than 1 degree of arc.

4.6.12 Low temperature exposure. The transmitter shall be subjected to a low temperature exposure test in accordance with MIL-STD-810, method 502. The transmitter shall be exposed to a temperature of -80 ºF ± 4 ºF (-62 ºC ± 2 ºC) for a minimum period of 48 hours after which it shall be operated throughout its entire range. It shall exhibit no erratic characteristics during operation at the -80 ºF ± 4 ºF (-62 ºC ± 2 ºC) temperature. The temperature shall then be raised to -65 ºF ± 4 ºF (-54 ºC ± 2 ºC) and the transmitter maintained at this temperature for a minimum of 24 hours. At the end of the 24-hour period, the transmitter shall be subjected to the scale error test specified in 4.6.2 and the friction error test specified in 4.6.3. The scale error and friction error shall not exceed the tolerances specified in table I.

4.6.13 High temperature exposure. The transmitter shall be placed in a chamber and subjected to a temperature of 446 ºF ± 9 ºF (230 ºC ± 5 ºC) for a period of 80 hours. At the end of this period, the transmitter shall be returned to room temperature for four hours after which it shall again be subjected to the scale error test specified in 4.6.2. The scale error shall not exceed the tolerances specified in table I for during and after high and low temperature exposure.

4.6.14 Altitude temperature. The transmitter shall be subjected to an absolute pressure equivalent to an altitude of 85,000 feet for a period of one hour at a temperature of -65 ºF ± 4 ºF (-54 ºC ± 2 ºC). While still at this pressure and temperature, the transmitter shall be subjected to the scale error test specified in 4.6.2. The scale error shall not exceed the tolerances specified in table I.

4.6.15 Electromagnetic interference. The transmitter shall be subjected to the electromagnetic interference test specified in MIL-STD-461.

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4.6.16 Salt fog. The salt fog test shall be conducted in accordance with MIL-STD-810, method 509, at room temperature except the exposure shall be for a continuous period of 48 hours. The transmitter shall be examined and there shall be no evidence of external corrosion or deterioration that would affect subsequent operation. The transmitters shall then meet the dielectric strength test specified in 4.6.5. Before the dielectric strength test is conducted, the external of the transmitter may be washed with distilled water and air dried.

4.6.17 Voltage and frequency. This test shall be conducted at room temperature. The transmitter shall be subjected to the scale error test specified in 4.6.2 and the friction error test specified in 4.6.3, during which the voltage and frequency shall be varied from 22 Vac to 30 Vac and 380 Hz to 420 Hz, or any combination thereof. The scale and friction errors shall not exceed those specified in table I.

4.6.18 Effect of lead length. This test shall be conducted at room temperature. The lead length between the transmitter and indicator synchro stators shall be changed from 5 to 300 feet of No. 20 AWG stranded copper wire. The indication shall not change more than 1 degree of arc.

4.6.19 Pressure bosses and connector strength. This test shall be conducted at room temperature and with the transmitter secured by the mounting flange. The pressure bosses and connector mountings shall be sufficiently rigid to withstand a force of 25 foot-pounds of bending movement applied in any direction perpendicular to the longitudinal axis of the transmitter without calibration shifting out of tolerance, distortion of the base plates, or evidence of damage to the pressure bosses, connector welds, or soldering. At the conclusion of this test, the transmitter shall be subjected to and meet the requirements of the case leakage test specified in 4.6.6.

4.6.20 Overpressurization tests.

4.6.20.1 Overpressure (750 psi). This test shall be conducted at room temperature. A pressure of 750 psi shall be applied to the transmitter for ten minutes. Not less than one hour following this test, the transmitter shall be subjected to the scale error test specified in 4.6.2 on increasing pressure only. The scale error shall not exceed the tolerances specified in table I for overpressure. Application of the 750 psi overpressure shall position the pointer of the indicating standard off the scale approximately midway between the 500-psi and 0-psi graduations as required for the overpressure stop in 3.3.1.8.

4.6.20.2 Overpressure (1,000 psi). The transmitter case shall be punctured or the seal shall be broken and a pressure of 1,000 psi shall be applied at the input pressure port. Leakage of the internal mechanism through the hole in the case shall be cause for rejection. The sample transmitter need not be checked for accuracy following this test. The transmitter shall then be disassembled and inspected for compliance with 3.3.1.7.1, 3.3.1.7.2, 3.3.1.9, and 3.3.1.9.1.

4.6.20.2 Case pressure release. The pressure blowout device shall be tested either under simulated increasing case pressure or by actual injection of pressure into the transmitter case. The device shall operate at a pressure of 90 ± 30 psi relative to the atmosphere.

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4.6.21 Damping. A 500 psi pressure shall be applied to the transmitter. A quick release valve shall be used to release the pressure so that the time interval for the pressure to decay from 500 psi to 100 psi shall be no more than 0.1 second. The corresponding time for the transmitter indication to change from 500 psi to 100 psi shall be 3 to 7 seconds. This test shall be conducted with oil, such as 1005 oil or equivalent, supplied to the pressure port of the transmitter, and the oil pressure and transmitter output monitored with a strip chart recorder. Neither restrictors nor a capillary line shall be used to connect the pressure input to the pressure element. This test shall be conducted three times at -65 ºF ± 4 ºF (-54 ºC ± 2 ºC). Following the damping test, the transmitter shall be subjected to the scale error test specified in 4.6.2 and the friction error test specified in 4.6.3.

4.6.22 Filling medium. The case shall be punctured or the filling tube cut on the transmitter case. Helium leakage shall be tested by means of a mass-spectrometer-type helium leak detector. Failure to detect helium leakage shall be cause for rejection.

4.6.23 Reliability test plan and report. Prior to starting the reliability test, the manufacturer shall submit a detailed reliability test plan for formal approval by the procuring activity. The manufacturer may refer to the requirements specified in test plan I-D of MIL-HDBK-781 as a guide for the reliability test plan. The manufacturer shall prepare a final report that summarizes the achieved reliability of the test samples. The report shall also contain test procedures, details of any malfunctions, and applicable charts and graphs showing failures/MTBF plotted against test time and other pertinent test data. The manufacturer may refer to the guidelines outlined in MIL-HDBK-831 for preparing the report.

4.6.23.1 Reliability test. The seven transmitters shall each accumulate 2,000 hours of test time by being subjected to a series of environmental test cycles. The transmitters shall be checked every 25 test cycles (250 hours) for scale error. The scale error of each transmitter shall be within the tolerances specified in table I for during and after reliability testing. Each test cycle is approximately 10 hours in duration and shall meet the following requirements:

a. The test cycle shall begin when rated power (26 Vac ± 2 Vac at 400 Hz ± 5 Hz) is applied to the transmitter and the temperature is at -65 °F (-54 °C). The temperature shall be cycled as follows: 4 hours at -65 °F (-54 °C), increase temperature to 329 °F (165 °C) at a rate of 9 °F per minute (5 °C per minute); 4 hours at 329 °F (165 °C), lower temperature to -65 °F (-54 °C) at a rate of 9 °F per minute (5 °C per minute), and repeat cycle. Prior to the temperature changing from low to high and high to low, the power shall be turned off for a period of 10 minutes during each 10-hour cycle.

b. Oil pressure shall be applied to the transmitter at a rate of 3 to 4 cycles per minute (cpm) from 0 to 80 ± 5 percent of full scale and return to 0 for the duration of the 2,000 hours of testing.

c. The transmitter shall be vibrated in accordance with MIL-STD-810C (10 March 1975), method 514, figure 514.2-2, curve H at 10g for 10 minutes out of every hour of power "ON" time. After every 25 test cycles the transmitter shall be reoriented 90 degrees so that another axis is tested.

4.6.24 Sensors matching. The transmitter adjustment shall be adjusted and locked for operation with the test indicator and the adjustment shall not be further disturbed for the balance of the sensor matching test. For each of the following conditions, the transmitter shall be tested and meet the scale error (sensor matching) at room temperature at the test points specified in

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table I for both up scale and down scale readings. The test point on the up scale readings shall be measured by bringing the input reading up to, but not exceeding, the specified test point. The down scale error shall be measured in the same manner except that the input reading value shall be brought down to, but not below, the specified test point.

4.6.24.1 20 psi vent pressure. A 20 psi pressure shall be applied to the vent port connection of the transmitter. Pressures shall be applied to the pressure port connection, which shall exceed that applied to the vent port connection by each of the test points listed in table I.

4.6.24.2 40 psi vent pressure. A 40 psi pressure shall be applied to the vent port connection of the transmitter. Pressures shall be applied to the pressure port connection, which shall exceed that applied to the vent port connection by each of the test points listed in table I.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contract personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general, or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The TRU-81/A transmitter covered by this specification is intended for use in transmitting the lubricating system oil pressure of aircraft engines by electrical signal to an indicator or indicators.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of individual documents referenced (see 2.2.1 and 2.3).
c. Electromagnetic interference requirements, if different (see 3.4.8).
d. Special test conditions, if any (see 4.2).
e. Sampling plan, A or B (see 4.5).
f. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in QPL-38394 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that

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they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

6.4 Subject term (key word) listing.

aircraft
differential
remote-indicating
signal
synchro

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notions. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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